OPPORTUNITY BRIEF · SALES AND PRESALES

Provide low-power sensor networks for monitoring gaseous emissions and pollutants at mining sites, as seen in Ogun East.

Natural resources × Emissions and environmental monitoring × Industrial sensors and instrumentation · OS132 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
60 is the world moving	**13** can we play, can we win	**MEDIUM** 0 named in the evidence	**€93.6m** modelled evidence · per year	**NOW** budgeted procurement within 1...	**L3** 8 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 2.0 published reference(s) in Natural resources, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

We can help mining operators in regions like Ogun East deploy low-power sensor networks to continuously monitor gaseous emissions and pollutants, enabling real-time environmental compliance and worker safety. This opportunity is immediate because environmental agencies are actively assessing such sites, and the technology is mature enough to deploy now.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€125m** €25.4m – €542m	**€93.6m** €18.9m – €405m	**€117k** €24k – €507k	modelled
Observed: tendered contract value, annualised	€28.0m	€28.0m	€35k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Natural resources (EU27_2020)	16,115 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Industrial sensors and instrumentation	13.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOTDPP	2021
Annual value of one engagement	€320k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	2,847 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. IoT for production processes Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract value is a configured assumption, not observed. Only 1 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (NG); the estimate covers the European footprint only. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes mining; PROXY — metal ores; PROXY — mining support; PROXY — oil and gas extraction; PROXY — other mining

Observed: tendered contract value, annualised

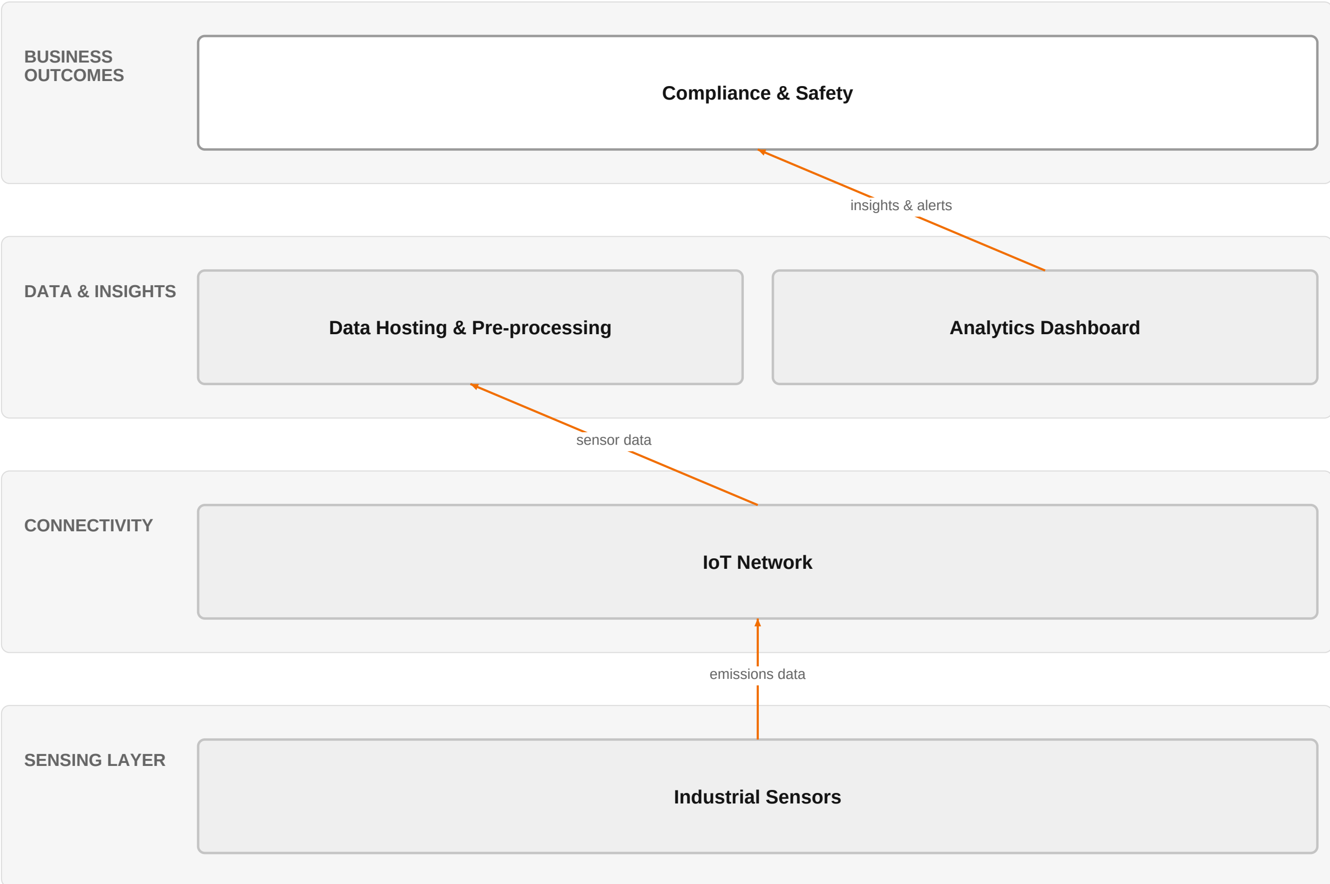
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€28.0m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-07-22

Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 7 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (NG); the estimate covers the European footprint only.

The solution, in one picture

Low-Power Emissions Monitoring Solution



Orange asset · Partner · Customer-owned · Third party / to be sourced

This diagram shows how sensor data flows from the field to actionable insights for compliance and safety.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Environmental agencies are actively assessing gaseous emissions and pollutants at mining sites, as seen in Ogun East, creating regulatory pressure for operators to monitor and report. Advances in industrial sensors and IoT enable low-power, continuous monitoring that was previously impractical. Additionally, research highlights the need for enhanced worker safety in hazardous mining environments through advanced sensors and IoT.

`Sources: SIG-9E90687CF1FE The Guardian Nigeria N 2026-08-16; SIG-97250BAF9D1E openalex.org 2025-10-07`

Who buys, and why now

The environmental compliance manager or site operations director at a mining company feels the pain of manual or infrequent emissions checks that risk non-compliance fines and worker safety incidents. The trigger is an environmental agency assessment like the one in Ogun East, which forces immediate action to demonstrate monitoring capability.

`Sources: SIG-9E90687CF1FE The Guardian Nigeria N 2026-08-16`

Why it is hot now — every claim bound to a dated source

- Environmental agencies are actively assessing gaseous emissions and pollutants at mining sites, creating a need for monitoring solutions. [SIG-9E90687CF1FE]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Digital experts to design and deploy a low-power sensor network, leveraging Industrial sensors and instrumentation. The engagement would start with a site assessment, then deploy sensors, and finally provide data hosting and pre-processing to deliver actionable insights to the customer.

Why Orange can win

Orange brings a combination of IoT expertise and digital transformation capabilities that pure sensor vendors lack. However, the asset list is thin on specific environmental monitoring offerings, so we must lean on our integration and data management strengths to differentiate.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Industrial sensors and instrumentation	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 4.7; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Bosch	Industrial / OT specialist	sells Industrial sensors and instrumentation; competes in OX: Smart Industries	structural	—
Honeywell	Industrial / OT specialist	sells Industrial sensors and instrumentation; active in Natural resources; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	sells Industrial sensors and instrumentation; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	sells Industrial sensors and instrumentation; active in Natural resources; competes in OX: Smart Industries	structural	—
Marlink	Satellite operator	active in Natural resources; competes in Connectivity Solutions	structural	—
Nokia	Network equipment vendor	active in Natural resources; competes in Connectivity Solutions, OX: Smart Industries	structural	—
SpaceX Starlink	Satellite operator	active in Natural resources; competes in Connectivity Solutions	structural	—
AWS	Hyperscaler	competes in OX: Smart Industries — also an Orange partner	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. How are you currently monitoring gaseous emissions and pollutants at your mining sites?
2. Have you been contacted by environmental agencies like OGEPA for an assessment?
3. What is your biggest pain point: compliance reporting, worker safety, or both?
4. Do you have existing connectivity infrastructure at the site, or would we need to provide it?
5. What is your budget and timeline for implementing a monitoring solution?

6 Who would be the primary decision-maker for this investment?

Objections you will meet

They will say	You can answer
We already have sensors from Siemens/Honeywell.	That's a solid foundation. Our value is in integrating those sensors with a data platform that gives you real-time alerts and compliance reports, which you may not have today.
This is too expensive.	We understand cost is a concern. Our solution is designed to be low-power and low-maintenance, reducing operational costs over time. We can also start with a pilot to demonstrate value.
We don't have the IT staff to manage this.	That's exactly why we offer a managed service. Our team handles the data hosting and pre-processing, so you get insights without needing to build an IT department.

Risks and unknowns

The main risk is that mining operators may prefer to buy sensors directly from established industrial vendors and handle integration themselves. Unknowns include the specific regulatory requirements in Ogun East and whether the customer already has a monitoring system in place.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief that differentiates Orange's offering by combining Industrial sensors and instrumentation (L3) with IoT and Digital experts (SUP) to deliver an end-to-end monitoring service, and identify the specific certifications or standards that would strengthen the proposal.
Sales	In a conversation with the environmental compliance lead at a mining operator in the Ogun East region, say: 'We can help you meet the growing scrutiny on gaseous emissions with a low-power sensor network that feeds data straight to your compliance team.'
Strategist / Innovator	Commission a deep-dive brief on the regulatory and operational drivers for emissions monitoring at mining sites, leveraging Orange Group researchers and IoT experts to map the sensor-to-platform architecture.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-FF96E1BC7BA5	2026-08-05	ted.europa.eu	1	Germany – Sensors – GE grunt, Sensorik, datenqualitatsgesicherter Betrieb, Datenhosting und-Vor...
SIG-1F577A1F9D8B	2026-06-09	ted.europa.eu	1	Austria – Sensors – Lieferung von Schwingungssensoren fur eine Carbonseil-Innovationssektion
SIG-B076E0F3616E	2026-05-20	ted.europa.eu	1	Italy – Sensors – DAC.0129.2025-Fornitura e posa di sistemi di monitoraggio per ponti e viadott...
SIG-0E742BED215C	2025-12-23	openalex.org	1	Improving Multivariate Time-Series Anomaly Detection in Industrial Sensor Networks Using Entrop...
SIG-97250BAF9D1E	2025-10-07	openalex.org	1	Investigating advanced technologies to enhance worker safety in hazardous mining environments: ...
SIG-9E90687CF1FE	2026-08-16	The Guardian Nigeria News	2	OGEPA assesses gaseous emissions, pollutants at Ogun East Mining Site - The Guardian Nigeria Ne...
SIG-4D1BCBC2C864	2026-07-14	Envirotech Online	2	Mercury emissions monitoring for environmental compliance - Envirotech Online
SIG-565963A66A79	2026-06-01	arxiv.org	3	IstGPT: LLM-based Anomaly Detection for Spatial-Temporal Graph in Industrial Systems

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:11+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.